

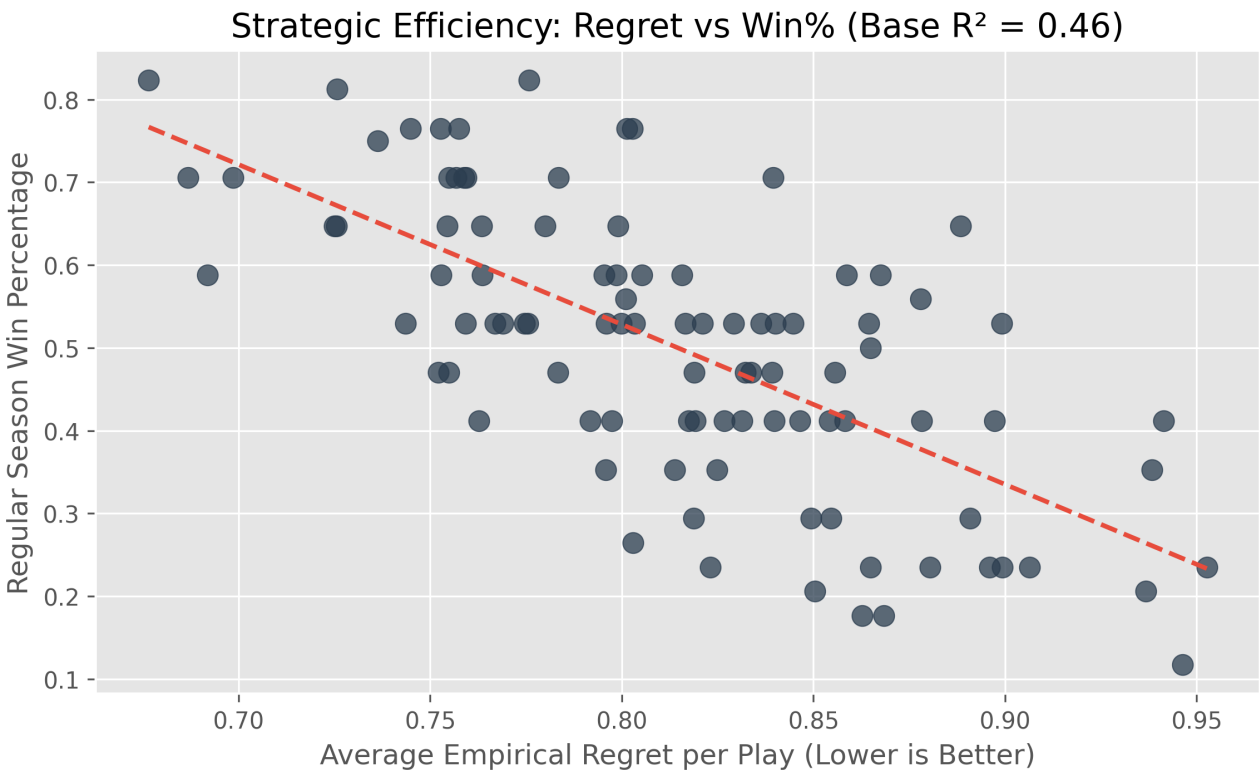
NFL AI Coach: Multivariate Contextual Bandit Analysis

1. Theoretical Methodology (Linear Bandit)

This methodology uses a Contextual Bandit framework. The AI Learner optimizes offensive play-calling against an Adversary (AI Defensive Coordinator) to accurately simulate counterfactual adjustments. To satisfy the theoretical constraint of conditionally sub-Gaussian noise, catastrophic execution outliers (turnovers) were excluded during the learner's regret simulation.

2. Multivariate Empirical Validation

While the pure strategy formulation (Base R-squared: 0.4630) proves that schematic regret strongly dictates winning, reality contains execution variance. By integrating player talent and turnovers back into a Multivariate OLS regression, the explanatory power leaps to a Multivariate R-squared of 0.5221. This successfully maps the entire equation: Strategy + Execution = Wins.



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